ISA5810 DM2026 Lab 1 — Master Notebook Questions

Fill in this file as you work through DM2026-Lab1-Master.ipynb. Submit this completed file converted to PDF together with the rest of your Lab 1 work.

Please do not change the style of this document.

Question numbers 1–10 follow the same order as the notebook (top to bottom).

Student Information

Please complete the fields below. For **English Name** and **Chinese Name**, you may fill in **only one** if you do not have the other.

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Instructions

These questions go with the Master Jupyter notebook (DM2026-Lab1-Master.ipynb). Each question is tied to a specific notebook section. Open the notebook, find **Question N (take home)**, and write your answer here.

We **strongly** suggest running the notebook to learn the lab, but the TAs will **not** audit whether you executed every cell to answer the questions in this document.

How to answer each question

1. Find the **notebook section** listed under the question (use the Table of Contents if needed).
2. Read the task; change parameters when asked (e.g. stop_words, thresholds, k, minSup).
3. Write a clear answer in the **Answer** space below each question.
4. Leave the question text unchanged; write only in the answer and disclosure fields.

Sources, honesty, and AI use

Acceptable vs. Unacceptable AI Usage: Generative AI should serve as an interactive tutor to guide your understanding, not as an automated solver completing the assignment for you.

- **Permitted (AI as Guide & Editor):**
  - Asking AI to explain unfamiliar theoretical concepts, discuss algorithmic trade-offs, or troubleshoot runtime errors in the notebook.
  - Polishing the grammar, phrasing, and readability of your own original, self-written drafts.

- **Prohibited (AI as Solver):**
  - Copying assignment questions directly into an AI tool to generate answers.
  - Using AI to fabricate experimental results or observations without executing the code in DM2026-Lab1-Master.ipynb.
  - Submitting machine-generated prose wholesale.

- **External material:** If you use websites, books, papers, or documentation beyond the notebook, cite them with **clickable links** (or DOI/ISBN when no URL exists).

- **When AI was used:** Your answer is acceptable if you (1) **disclose** AI use and (2) include **at least one external link** that validates the information in your answer (official docs, textbook, paper, or reputable tutorial — not the AI chat alone as the only source).

- **When AI was not used:** You may write "No" under **AI used** or leave it blank only if you truly did not use AI for that question. References are still encouraged if you looked anything up.

Under each question, complete:

- **AI used for this answer:** Yes / No
- **If yes, how:** (e.g. brainstorming, grammar, concept explanation)
- **External references (links):** *(required when AI = Yes; encouraged otherwise)*

Write in English. Keep answers concise and specific.

Questions

Question 1: Data Filtering & Domain Relevance

Notebook: Section 3.2 — Question 1 (take home)

Task: Filter the dataframe **X** for the category **sci.med** and check its length.

Questions:

1. How many documents are in this specific category?
2. From a data mining perspective, why is isolating specific categories a crucial first step before applying algorithms like frequent pattern mining?

Answer:

1. 594 documents
2. If you run frequent pattern mining without filtering for the right domain data first, the results can be misleading. General words tend to show up across different domains, which overly inflates their importance. Also, if there is a massive amount of data from one specific category, the most common terms might end up being dominated by that single domain.

AI used for this answer: Yes

If yes, how: concept explanation, grammar, translation

External references (links):

https://pandas.pydata.org/docs/user_guide/indexing.html#boolean-indexing

https://en.wikipedia.org/wiki/Association_rule_learning

Question 2: Handling Missing Data

Notebook: Section 4.1 — Question 2 (take home) (after the **dropna()** demonstration)

Task: In Section 4.1, a dummy record with a missing value is injected and then removed using the **dropna()** function.

Question: If 40% of documents were missing the **category_name** label, why would simply dropping these rows (with **dropna()**) be a flawed data mining strategy? Name one objective alternative so the text data can still be utilized.

Answer:

Dropping 40% of the documents with dropna() would throw away a large amount of useful text data, so the remaining dataset would be much smaller and might no

longer represent all the categories well. A simple objective alternative is to use the category column: since each integer in category corresponds to exactly one category_name, the missing names can be filled back in from this mapping. If the category number were also missing, we could use the text features (e.g., the document-term matrix) to cluster the documents and assign each unlabeled document to the group it is most similar to.

AI used for this answer: Yes

If yes, how: concept explanation, grammar, translation

External references (links):

https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.dropna.html

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Question 3: Data Sampling Risks

Notebook: Section 5.1 — Question 3 (take home)

Task: In Section 5.1, the code uses **X_sample = X.sample(n=1000)** to create a subset. Compare the bar chart of the original category distribution to the sampled distribution.

Questions:

1. Do the exact category proportions perfectly match the original?
2. What is the specific risk of simple random sampling on a highly imbalanced dataset?

Answer:

1. No, the proportions do not match exactly, but the overall trend is similar. For example, comp.graphics is 25.9% of the original data but 27.8% of the sample, while soc.religion.christian drops from 26.5% to 24.8%.
2. If the dataset is highly imbalanced, simple random sampling might not select any documents from a very small category at all, so that category could be completely missing from the sample.

AI used for this answer: Yes

If yes, how: concept explanation, grammar, translation

External references (links):

https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.sample.html

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Question 4: Text Vectorization and Matrix Sparsity

Notebook: Section 5.3 — Question 4 (take home)

Task: In Section 5.3, the **CountVectorizer** converts text into a document-term matrix using **stop_words='english'**. Change this parameter to **stop_words=None** and re-run the cell to check the shape of **X_counts**.

Questions:

1. Does the total number of features (columns) increase or decrease?
2. Since stopwords are highly common words that appear in almost all documents, what effect does adding them back have on the overall sparsity (the percentage of zeros) of the matrix?

Answer:

1. The number of features increases, from 35,482 to 35,788 (306 more columns), because the stopwords that were removed with stop_words='english' are now included in the vocabulary.
2. Since stopwords appear in almost every document, their columns are mostly non-zero, so adding them back reduces the percentage of zeros and makes the matrix slightly denser. However, the matrix is still extremely sparse (about 99.5% zeros), because the stopwords are only 306 of the 35,788 columns.

AI used for this answer: Yes

If yes, how: concept explanation, grammar, translation

External references (links): https://scikit-learn.org/stable/modules/generated/sklearn.feature_extraction.text.CountVectorizer.html

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Question 5: Feature Selection — Variance Thresholding

Notebook: Section 5.3.1 — Question 5 (take home)

Task: In the variance filtering section, re-run **variance_filter** using threshold values **threshold = 0.0, threshold = 0.01, and threshold = 0.05**. Record the number of features marked as **keep == True**.

Questions:

1. What happens to the total size of the vocabulary as the variance threshold increases from 0.0 to 0.05?

2. In terms of document frequency, what kinds of words have variance close to zero and are eliminated first?
3. Why is eliminating near-zero variance features considered a safe way to reduce dimensionality without harming model accuracy?

Answer:

Threshold	Number of terms kept (keep == True)
0.0	35482
0.01	5089
0.05	1491

1. As the threshold increases from 0.0 to 0.05, the vocabulary becomes smaller: 35,482 terms are kept at 0.0, 5,089 at 0.01, and 1,491 at 0.05.
2. Rare words have variance close to zero. Even when they appear, they appear in only a few documents and with very low frequency, so their counts are almost always 0. These words (e.g., zumder, kuala, 0000) are eliminated first.Why is eliminating near-zero variance features considered a safe way to reduce dimensionality without harming model accuracy?
3. Because these rare words are 0 in almost every document, they do not help distinguish between categories, so removing them reduces the number of features without hurting classification.

AI used for this answer: Yes

If yes, how: concept explanation, grammar, translation

External references (links): https://scikit-learn.org/stable/modules/generated/sklearn.feature_selection.VarianceThreshold.html

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Question 6: Feature Correlation — Linear (Pearson) vs. Monotonic (Spearman)

Notebook: Section 5.3.2 — Question 6 (take home)

Task: Run the correlation filter against the binary indicator for the **comp.graphics** class. Inspect the top 10 features sorted by **pearson_r** and compare them to the top 10 features sorted by **spearman_r**.

Questions:

1. Are the top-correlated words generic stop words or domain-specific terminology for graphics?
2. What is the mathematical difference in what **Pearson** correlation measures versus what **Spearman** rank correlation measures?
3. In what specific scenario would a word have a high **Spearman** rank correlation with a topic, but a significantly lower **Pearson** correlation?

Answer:

1. Based on Pearson, the top words are mostly common words (e.g., thanks, hi, writes), while based on Spearman, the top words are mostly domain-specific graphics terms (e.g., graphics, image, 3d, files).
2. Pearson uses the raw counts of a word and checks whether the counts increase linearly with the class label. Spearman first converts the counts into ranks and then checks whether they increase together.
3. When a word appears a very large number of times in a few documents, while in most documents of that topic it appears only once or twice. These extreme counts make the relationship far from linear, so Pearson becomes low, but after converting to ranks they are just the top ranks, so Spearman stays high. For example, “graphics” has Spearman 0.52 but Pearson only 0.18.

AI used for this answer: Yes

If yes, how: concept explanation, grammar, translation

External references (links):
https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.spearmanr.html

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Question 7: Frequent Pattern Mining — Algorithm & Parameter Dynamics

Notebook: Section 5.4.2 — Question 7 (take home)

Task: In the Frequent Pattern Mining section, run **Top-K FAE** with **k = 100 and k = 500**. Then, run **MaxFPGrowth** using minimum support thresholds **minSup=9 and minSup= 3**.

Questions:

1. How does decreasing **minSup** (from 9 to 3) in **MaxFPGrowth** affect the execution runtime and the total number of discovered patterns?
2. How does the parameter k in **Top-K FAE** differ fundamentally from **minSup** in **MaxFPGrowth** when deciding which patterns to keep?
3. Why are "maximal" frequent patterns structurally more concise than standard frequent patterns?

Answer (1) — Runtime and pattern count when lowering minSup:

When minSup decreases from 9 to 3, the number of patterns increases (710 → 9547) and the runtime becomes longer (0.046s → 0.888s). This is because word combinations that appear in only 3 to 8 documents, which were excluded before, now also count as frequent and have to be considered.

Answer (2) — Top-K k vs MaxFPGrowth minSup:

Changing k directly changes how many patterns are selected: k fixes the number of patterns (exactly 100 or 500), and the support threshold changes by itself (the lowest support dropped from 21 to 16). minSup in MaxFPGrowth is the opposite: the threshold is fixed (9 or 3), and the number of patterns changes (710 vs 9547).

Answer (3) — Maximal vs standard frequent patterns:

If a larger combination is frequent, every smaller combination inside it must also be frequent. So the smaller combinations do not need to be written out, which naturally makes the list more concise.

AI used for this answer: Yes

If yes, how: concept explanation, grammar, translation

External references (links): https://github.com/UdayLab/PAMI

Question 8: Dimensionality Reduction — Motivation

Notebook: Section 5.5 — Question 8 (take home) (before PCA/t-SNE code)

Task: After text vectorization, the **X_counts** matrix creates tens of thousands of distinct features (words).

Questions:

1. If we want to visually map how these documents group by category on a scatter plot, why is it mathematically and visually impossible to plot the raw **X_counts** matrix directly?
2. What specific category of data mining algorithms (referenced in the helper files) must be applied first?

Answer:

1. Each document in X_counts is a point with 35482 dimensions (one per word), but a scatter plot can only show 2 dimensions (or 3 in a 3D plot). So the raw X_counts matrix cannot be plotted directly.
2. Dimensionality reduction algorithms must be applied first, such as PCA, t-SNE, or UMAP, to reduce the data to 2 or 3 dimensions.

AI used for this answer: No

If yes, how:

External references (links):

Question 9: Dimensionality Reduction — Linear (PCA) vs. Non-linear Manifolds (t-SNE/UMAP)

Notebook: Section 5.5 — Question 9 (take home) (after 2D/3D plots)

Task: In the 2D/3D Dimensionality Reduction section, compare the scatter plot projection generated by **PCA** against that generated by **t-SNE** or **UMAP**.

Questions:

1. Which method **(PCA or t-SNE/UMAP**) shows distinct, separated cluster islands for the 4 document categories, and which shows heavy overlap around the center?
2. Why does **PCA** struggle to separate high-dimensional text categories compared to **t-SNE** or **UMAP**?
3. Why must data analysts be careful not to over-interpret the physical distance between separate clusters in a **t-SNE** plot?

Answer:

1. t-SNE/UMAP shows more separated clusters for the 4 categories, while PCA shows heavy overlap, with most points squeezed into one area.
2. Because the data has very high dimensions (35482 words), and PCA is a linear compression. It is unlikely that a linear projection down to only 2 dimensions can separate all the categories.
3. Because t-SNE is a non-linear compression that only keeps nearby points close. Two clusters that look far apart on the plot may actually be close in the original data, like the path takes a detour on the plot. So the distance between separate clusters does not show how different they really are.

AI used for this answer: Yes

If yes, how: concept explanation, grammar, translation

External references (links): https://scikit-learn.org/stable/modules/generated/sklearn.manifold.TSNE.html

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Question 10: One-Hot Binarization

Notebook: Section 5.6 — Question 10 (take home)

Task: In Section 3.1, examine **X['category']** (integers 0, 1, 2, 3). Then look at the output of **LabelBinarizer** in Section 5.6, where each category is converted into a binary vector (e.g., [1, 0, 0, 0]).

Questions:

1. How many binary columns are created to represent the 4 categories?
2. If an algorithm (like linear regression or KNN) is given the raw integer column **(0, 1, 2, 3)** as input, what false mathematical assumption does it make about the categories?
3. How does one-hot binarization eliminate this false assumption?

Answer:

1. 4 binary columns are created, one for each category.
2. The algorithm would treat the categories as ordinal data, as if they had an order (3 is larger than 1) and meaningful distances (1 is closer to 2 than to 3). But the 4 newsgroup categories have no real order.
3. After one-hot binarization, every two categories have the same distance from each other, so there is no order and no category is closer to another.

AI used for this answer: Yes

If yes, how: concept explanation, grammar, translation

External references (links): https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.LabelBinarizer.html

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